



Post-hurricane recovery and long-term viability of the Alabama beach mouse



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ARTICLE INFO

Article history:

Received 10 October 2013

Received in revised form 9 July 2014

Accepted 12 July 2014

Keywords:

Beach mouse

Peromyscus polionotus ammobates

Occupancy

Viability

Spatial model

Hurricane

Habitat loss

Bayesian

ABSTRACT

The spatial configuration of habitat and the frequency of disturbances through time could have interacting effects on population viability. With this in mind, we assessed the rate of post-hurricane recovery and long term viability of the Alabama beach mouse (*Peromyscus polionotus ammobates*). We collected detection/nondetection data across the range of the Alabama beach mouse for four years subsequent to hurricanes Ivan and Katrina, and then fit a dynamic patch occupancy model to these data using hierarchical Bayesian methods. We converted remotely sensed data into habitat classes, and then mapped site survival probability over the entire range of the species. These estimates were then used to parameterize a spatially-explicit population viability analysis. Our occupancy modeling demonstrates that the probability of patch occupancy increased from 0.16 to 0.67, with occupancy in the surveyed patches approaching an asymptote by the third post-hurricane summer. The viability analysis suggests that extinction probability increases nonlinearly with monotonic increases in both habitat loss and hurricane frequency. Extinction risk is sensitive to the regularity of catastrophic hurricanes; consecutive hurricanes dramatically increase extinction risk, further suggesting that the effect of global climate change on hurricane regimes may have a large effect on the probability of long-term persistence. A mild interaction between habitat loss and hurricane frequency occurs when extinction probability is relatively low, but disappears as extinction probability increases. Our results indicate that extinction risk is a complex function of multiple interacting drivers.

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1. Introduction

Extinction risk is a frequently used metric in environmental management and decision making. Often that risk is expected to be a function of one, overwhelmingly important, driver such as a form of major disturbances or habitat loss, or some biotic issue such as invasive species or disease. However, extinctions may ultimately be the result of multiple factors acting in a multiplicative or synergistic fashion, which complicates assessment of persistence times (e.g., Vuilleumier et al., 2007; Brook et al., 2008). Classic examples include specialized butterflies with shrinking ranges impacted by fire (e.g., Swengel, 1996; Charrette et al., 2006). Species with reduced ranges due to habitat loss tend to have smaller population sizes, and therefore face increased risk from disturbances than larger populations. We modeled anthropogenic habitat loss and the frequency of major disturbances that may be a product of global climate change in order to better understand this

synergy on the risk of extinction of an endangered species. The rate of post-disturbance population recovery is a critical uncertainty in such assessments. The Alabama beach mouse (*Peromyscus polionotus ammobates*) provided an opportunity for us to obtain empirical estimates of the rate of post-hurricane recovery.

Beach mice along the northeastern coast of the Gulf of Mexico are a group of endangered subspecies that face continued loss of habitat due to anthropogenic development. They also face severe hurricane disturbances that may be increasing in both frequency and intensity (Emanuel, 2005; Trenberth, 2005; Webster et al., 2005). These subspecies prefer lightly vegetated, low primary dunes that begin immediately behind the high-tide surf zone. To a much lesser degree, they also use the scrub dune habitat immediately inland of the primary dune field which has a higher elevation and is dominated by woody shrubs. Densities of mice in the scrub are normally much less than in the primary dunes (Pries et al., 2009). These lower densities in the scrub have justified substantial conversion of the scrub into condominiums, private residents, golf courses, and other developments that are by and large not compatible with beach mouse productivity while primary

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dunes have been, to a large extent protected from development and anthropogenic disturbances.

Yet, it is these protected primary dunes that are most susceptible to hurricanes, and in 2004 and again in 2005, the entire historic range of Alabama Beach Mouse was hit by hurricanes Ivan and then Katrina, eliminating almost all of the primary dunes and heavily damaging the scrub habitat. Overall, approximately 90% of Alabama beach mouse habitat was suspected to have been significantly damaged or destroyed by these hurricanes (USFWS, 2005). As a species, beach mice have survived hurricanes by persisting in low densities in the remains of scrub habitat until the primary dunes reform and become inhabitable, which can be 5–10 yrs. However, once extensive tracts of scrub habitat have now been largely developed. Beginning in the 1950s, but with much greater losses of habitat in the late 1980s and 1990s, scrub habitat has been reduced to small, isolated patches scattered along the landward borders of the beach mouse's range.

While other factors like predation (Falcu and Danielson, 2013a) and competition (Falcu and Danielson, 2013b) may play an important role in the population dynamics of beach mice, we focus our model only on the effects of habitat loss and hurricanes because these were cited as causes of decline when *P. p. ammobates* was granted federal protection under the Endangered Species Act. The effects of hurricanes on beach mouse population density (Swilling et al., 1998) and spatial distribution (Pries et al., 2009) have already been documented. Here, we use occupancy modeling to estimate of the rate of spatial recovery of the Alabama beach mouse following the catastrophic hurricanes of 2004 and 2005. Furthermore, population viability analyses (PVA) of the Alabama beach mouse have already been performed (Oli et al., 2000; Traylor-Holzer, 2005), but here we develop a spatially-explicit PVA parameterized with occupancy estimates to examine whether the risk of extinction from more frequent hurricanes is expected to be exacerbated by additional habitat loss.

2. Methods

Techniques for simulating occupancy dynamics of populations on binary landscapes of suitable and unsuitable habitat are relatively well-developed (Sjögren-Gulve and Hanski, 2000; Moilanen, 2004). However, many species do not occur on binary landscapes, and the importance of marginal habitats on population dynamics is well documented (Weigand et al., 2005; Falcu and Danielson, 2011). Occupancy models can be fit to data that were collected from investigator-defined sampling units, and covariates can be used to infer occupancy at un-surveyed locations (Royle and Dorazio p. 89). A challenge to inference and simulation of occupancy models when patches are not discrete concerns assumptions of the degree of spatial independence of site colonization and site survival. We address this issue by relating spatial variation in survival probabilities to local habitat conditions derived from remotely-sensed data, grouping regions of spatially-dependent survival, and then finding a colonization function that results in empirically observed occupancy rates at locations where detection/non-detection data were collected.

2.1. Beach mouse surveys

We conducted detection/non-detection surveys for the Alabama beach mouse every summer from 2006 through 2009. All survey sites were located in or to the west of the Perdue Unit of the Bon Secour National Wildlife Refuge (Fig. 1), where commercial and residential developments have had the greatest impact on habitat connectivity. Survey sites were either along public right-of-ways or on privately held property where access had been granted. The

number of sites falling into the latter category increased throughout the duration of the study because the U.S. Fish and Wildlife Service, while promulgating the federal Endangered Species Act, included a monitoring-access clause in the permits it issued to landowners applying for permission to modify their parcel's landscape. During the summer of 2006, we were able to survey just 31 sites; by summer of 2009, 61 sites were included in our surveys.

Surveys were conducted with Sherman live traps. Traps were deployed at a density of approximately one per 225 m², with the total number of traps per site ranging from four to 32, depending on site size. Traps were set in the evening and then checked for beach mice in the early morning. Individuals were then released at the point of capture. Each site was surveyed once per year over five consecutive nights, with rare exceptions due to inclement weather.

2.2. Dynamic occupancy model

We adopted a hierarchical Bayesian approach to occupancy modeling because the resulting posterior probability distributions of latent process parameters can be easily used to simulate uncertainty in population viability analysis (Wade, 2002). The hierarchical approach to fitting dynamic occupancy models (Royle and Kéry, 2007; Royle and Dorazio, 2008; Link and Barker, 2010) to data requires two submodels: A state process submodel describes how occupancy status changes through time, and an observation submodel describes imperfection in the observation of that process. The observation model is important because the focal species may be present yet remain undetected during a survey. The observation model describes this uncertainty:

$$Y_{j,i,t} | Z_{i,t} \sim \text{Bernoulli}(p_{t,Z_{i,t}}) \quad \text{for} \\ j = 1, 2, \dots, 5 \text{ temporal replicate surveys,} \\ i = 1, 2, \dots, 61 \text{ survey sites, and} \\ t = 1, 2, 3, 4 \text{ years of observations}$$

where $y \in \{0, 1\}$ for nondetection and detection, respectively and $z \in \{0, 1\}$ for absence and presence, respectively.

The state model describes the ecological processes that drive occupancy dynamics. Following Royle and Kéry (2007) and Royle and Dorazio (2008), we use an auto-logistic formulation of change in occupancy through time because the probability of site occupancy at a given time, ψ_t , depends on whether or not the site is occupied at time $t - 1$. The state model for the first year that a site was sampled is simply:

$$Z_{i,1} \sim \text{Bernoulli}(\psi_1) \quad \text{for } i = 1, 2, \dots, 61 \text{ sites.}$$

The true occupancy status (z) of sites during subsequent years is the net result of interannual survival (Φ) and colonization (γ) probabilities:

$$Z_{i,t} | Z_{i,t-1} \sim \text{Bernoulli}(\pi_{i,t}) \quad \text{for } t = 2, 3, 4$$

where

$$\pi_{i,t} = \gamma_{t-1}(1 - Z_{i,t-1}) + \Phi_{t-1}Z_{i,t-1} \quad (1)$$

Hence, the probability of occupancy at time t depends on either colonization (γ) or survival (Φ), depending on whether the site was previously unoccupied ($Z_{i,t-1} = 0$) or occupied ($Z_{i,t-1} = 1$), respectively. Covariate coefficients (β) can then be estimated from the equation:

$$\text{logit}(\pi_{i,t}) = \gamma_{t-1}(1 - Z_{i,t-1}) + \Phi_{t-1}Z_{i,t-1} + \omega_1\beta_1\text{Cov}1_{i,t-1} \\ + \omega_2\beta_2\text{Cov}2_{i,t-1} + \dots + \omega_4\beta_4\text{Cov}n_{i,t-1} \quad (2)$$

Note that all covariates in Eq. (2) are placed on survival ($Z_{i,t-1}$) but not colonization ($1 - Z_{i,t-1}$) because we are interested in using the covariates to predict the former but not the latter. Each

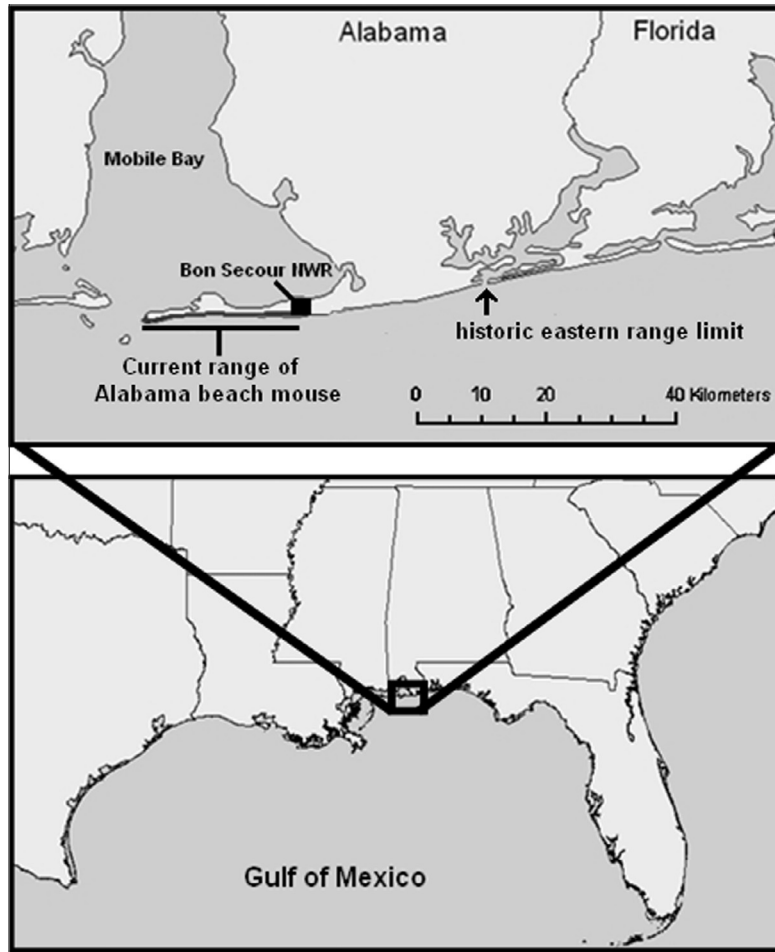


Fig. 1. Current and historical range of the Alabama beach mouse (*Peromyscus polionotus ammobates*).

posterior element of the covariate coefficient, β_k ($k = 1, 2, 3, 4$) is multiplied by the corresponding posterior element of a binary inclusion parameter, ω_k . The binary inclusion variables (ω_k) take on values of either 1 or 0 to indicate inclusion and exclusion, respectively, of the associated covariate in the model. If a given covariate contains little information, the associated inclusion variable will frequently evaluate to 0, causing the associated covariate to be dropped from Eq. (2). Thus, the inclusion variables function as covariate weights that are used to perform model averaging (Kuo and Mallick, 1998; Royle and Dorazio, 2008, Sections 2.5.3 and 3.4.3).

The hierarchical formulation of the model readily yields to Bayesian analysis. We specify our *a priori* neutrality in the values of p , Φ , and γ by assigning equally probable values in the range [0 1] for each parameter by using a uniform density distribution, i.e. $[p, \Phi, \gamma] \sim \text{Uniform}(0, 1)$. Similarly, prior to examining data, we used relatively noninformative prior values of β , but since these values are not probabilities, we define this with a normal distribution centered on 0 with moderately large standard deviation: $\beta \sim \text{Normal}(0, 12.5)$. Larger standard deviations caused convergence problems, but since all covariates were standardized this still represents a moderately vague prior. Neutrality toward whether or not a given covariate meaningfully contributes to predictions of the state variables was modeled: $\omega \sim \text{Bernoulli}(0.5)$, and sensitivity to this assumption was checked by rerunning the analysis with $\omega \sim \text{Bernoulli}(0.1)$.

The joint posterior probability density is given by the hierarchical factorization:

$$[y, z, p, \Phi, \gamma, \beta, \omega] = f(y|z, p)g(z|\Phi, \gamma, \beta, \omega)h(p, \Phi, \gamma, \beta, \omega), \quad (3)$$

where f is the distribution of the data given the observation model, g is the distribution of the true occupancy state given the parameter values in the process model, and h is the prior distribution of all model parameters. The joint posterior probability density was evaluated with the Gibbs sampler implemented in WinBugs (2009) and accessed through MATLAB (2009) via Matbugs (Murphy and Mahdavian, 2005). Inferences are based on a parameter's marginal posterior distribution, which may be obtained by simply ignoring the other parameters in a sample draw from the joint posterior. This draw is an implicit integration over all other parameters and, therefore, includes the uncertainty in estimating those parameters.

We performed two numerical evaluations of dynamic occupancy models. The first focuses on the changes in occupancy probability through time. This analysis does not include site-level covariates. Furthermore, this analysis uses only those data from sites that were sampled all four years ($N = 31$). Since the 30 additional sites that became available for sampling after the first year had a significantly lower probability of containing beach mice than the original 31 sites, estimates of occupancy probability through time would be biased downwards as more and more sites with low probability of occupancy are included in the analysis. The canonical parameters of this dynamic occupancy model are the initial occupancy (ψ_1), annual survival (Φ_t), and the annual colonization rate (γ_t). Occupancy in any given year subsequent to the first is calculated recursively

$$\psi_t = \psi_{t-1}\Phi_{t-1} + (1 - \psi_{t-1})\gamma_{t-1} \quad (4)$$

when inference to all possible sites is desired. We report the “finite sample” occupancy which is simply the mean number of sites occupied, $\frac{1}{31} \sum_{i=1}^{31} z_{i,t}$, during $t = 1, 2, 3, 4$ years after hurricanes. We then used nonlinear regression to estimate parameters of the model:

$$\psi = \frac{\beta_0}{1 + \beta_1 e^{-\beta_2 \text{years}}} \quad (5)$$

using the estimates of occupancy estimates.

We also report the equilibrium occupancy probability, which is a fundamental parameter of metapopulations (Hanski, 1994). Equilibrium occupancy is the anticipated long-term proportion of sites occupied when colonization and survival remain constant:

$$\psi_t^* = \frac{\gamma_t}{\gamma_t + (1 - \phi_t)}. \quad (6)$$

We also report the finite sample turnover, τ , which is the probability that an occupied site is a newly occupied one (Nichols et al., 1998). Thus, $\tau = \Pr(z_{t-1} = 0 | z_t = 1)$ and is computed:

$$\tau_t = \frac{\sum_{i=1}^{31} (1 - z_{i,t-1}) z_{i,t}}{\sum_{i=1}^{31} (1 - z_{i,t})}. \quad (7)$$

Finally, we also report the finite sample rate of metapopulation growth rate, λ , which is analogous to the well known population growth rate estimator: $\frac{N_{t+1}}{N_t}$. Here,

$$\lambda_t = \frac{\sum_{i=1}^{31} z_{i,t+1}}{\sum_{i=1}^{31} z_{i,t}} \quad (8)$$

Our second numerical evaluation of the dynamic occupancy model focused on the effects of site-level covariates on the probabilities of survival. This analysis used data from all 61 sites. We used estimates of the covariate coefficients (β) and their respective weights (ω) to create maps of survival probability throughout the entire range of the Alabama beach mouse. We used Eq. (2) to estimate the effects of four covariates on survival probability. We selected covariates that were biologically relevant and readily measurable across the entire range of the Alabama beach mouse because our goal was to map survival probabilities at this scale. We, therefore, chose covariates derived from remotely sensed data. We digitally centered a $167 \text{ m} \times 167 \text{ m}$ cell, which approximately matches the average size of a survey site, on all 61 survey sites and computed the following covariate information for each cell: (i) the fraction of the cell covered in sand, (ii) the fraction of the cell that is covered by something other than sand or vegetation, (iii) an index of homogeneity of color intensities within the cell, and (iv) the average elevation above sea level of the cell.

Beach mice are granivorous and make burrows in the sand. Thus, beach mice should respond to a sand-vegetation habitat gradient. Anything that is not sand or vegetation is likely to be unusable to beach mice (houses, roads, parking lots, standing water). If points within an area are classified as one of these three classes (sand, vegetation, other unusable habitat), then the area's habitat can be characterized with the proportions of just two classes because the proportions sum to 1. Both (i) and (ii) above were computed from color aerial photographs with 15 cm^2 pixel sizes taken in 2001. To do this, we first created a training dataset of pixels that were visually classified (50 sand, 112 vegetation, and 172 “other”). Next, we used an artificial neural network to discriminate among the three classes in the training data. The neural network was a three-layer, feed-forward, back-propagation network, with 20 nodes in the hidden layer. The back-propagation of errors was conducted with a hyperbolic tangent sigmoid transfer function (Vogl et al., 1988). The performance of the trained neural network was assessed by withholding 20% of the training dataset from each pixel class during network development. These validation data were subsequently classified by the network, and the assigned

classes were compared to known classes to estimate the frequency of misclassification. The trained neural network was then used to classify every pixel in the study area as sand, vegetation, or “other” based on the intensity of the pixels' red, green, and blue values. We then summed the number of pixels classified by the neural network as sand and “other” within each $167 \text{ m} \times 167 \text{ m}$ cell.

Homogeneity of color intensities within a cell (iii above) provides a description of feature complexity. A cell with intermixed sand and vegetation will have less homogeneity than a cell with sand and vegetation that are each clumped together. To compute homogeneity, the image is first rendered to grayscale with eight brightness classes. Next, an 8×8 matrix is constructed, with rows and columns corresponding to brightness classes. The elements of this matrix give the number of pixels of a given brightness (row) that are horizontally adjacent to pixels of other brightness values (column). The covariate is the sum of the squared elements in this matrix, which provides an index of how evenly brightness is distributed within the cell. Finally, the average elevation of a cell (iv above) is simply the average of all constituent $3 \text{ m} \times 3 \text{ m}$ LIDAR elevation readings. All aerial photography and LIDAR data were obtained from the U.S. Fish and Wildlife Service office in Daphne, Alabama.

2.3. Population simulations

Once survival covariate coefficients had been estimated, we computed survival for all $167 \text{ m} \times 167 \text{ m}$ cells covering the recent range of the Alabama beach mouse (ca. $27 \text{ km} \times 0.6 \text{ km}$). We manually excluded cells from the survival map that we considered to be uninhabitable. Our final survival map contains 398 cells (Fig. 2). Since Eq. (2) contains a time-varying intercept for survival, we were able to create three survival maps, each corresponding to one of the three interannual intervals of our survey (survival from 2006 to 2007, 2007 to 2008, and 2008 to 2009).

Adjacent cells are likely to have similar survival probabilities because of spatial autocorrelation of habitat (Fig. 2). However, cell size ($167 \text{ m} \times 167 \text{ m}$) was chosen to match the approximate area of survey sites, and does not necessarily match the spatial scale where the survival process can be considered independent of neighboring areas. A computationally simple way of simulating survival is to compare the estimated survival probability in a given location to uniformly distributed random numbers on the interval [0 1]. If the randomly drawn number is less than the survival probability, then a survival event is recorded, otherwise an extinction event is recorded. Rather than drawing a unique, uniform random number for all cells to simulate survival independently in all cells, we instead draw a single uniformly distributed random number and use that number to evaluate survival in a group of cells that are thought to operate dependently. This approach preserves information about which cells are more likely to remain occupied from one time step to the next (i.e. survive) based on habitat characteristics, but does not assume that the survival process happens to match the size of survey sites, which were determined by property boundaries. We selected 9 groups where survival was modeled independently of other groups (Fig. 2a). These groupings reflect differences in elevation, habitat and development.

Occupancy is the result of colonization and survival rates. Thus, if survival and occupancy are given, then colonization can be inferred. To identify a spatially explicit colonization function during model development, we held our survival algorithm constant and then gradually increased colonization ability until simulated occupancy at cells corresponding to empirical survey sites matched the empirical occupancy. However, since we observed a single post-hurricane recovery event, we selected a colonization function that produced additional variation in occupancy. The difference between the estimated and simulated variance in

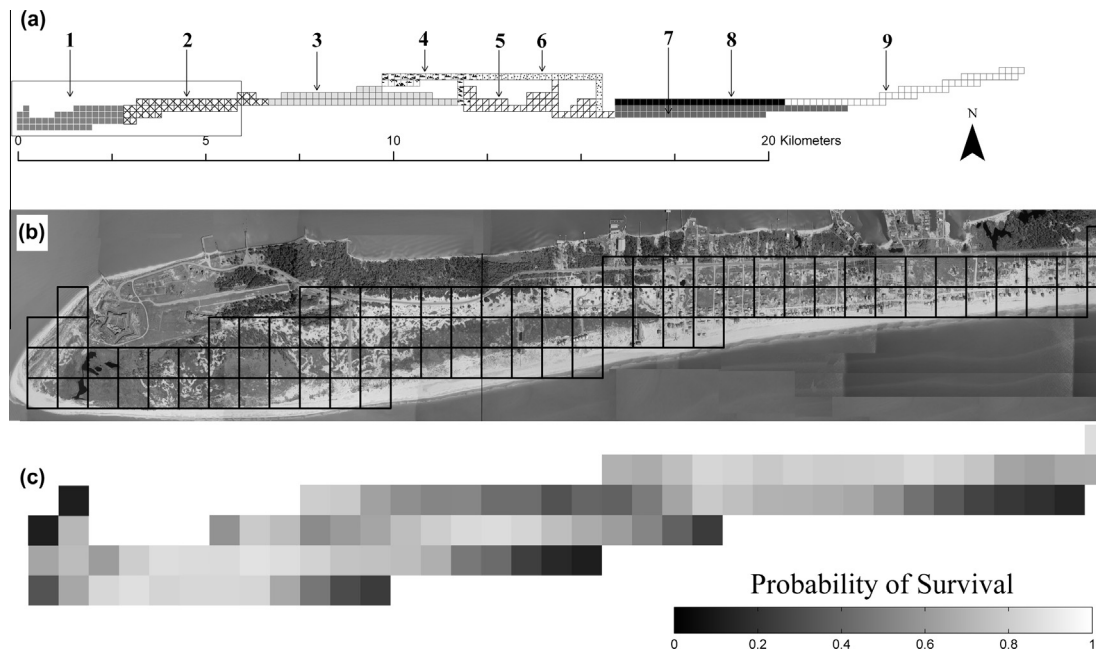


Fig. 2. Simulated landscape composed of 398 cells distributed over the range of the *Peromyscus polionotus ammobates* (a). Each cell is 167 m by 167 m. Different regions of the landscape, indicated by color and number, indicate groups of cells that experience similar inter-annual survival and hurricane survival. Rectangle indicates area given in panels (b) and (c). The western tip of the Ft. Morgan Peninsula is a mixture of sandy beaches, dunes, vegetation, and development (b). The estimated probability of survival after the third year post-hurricane in the on the western tip of the Ft. Morgan Peninsula (c). Note that the simulated landscape excludes areas where survival is likely to be very low and habitat is unlike the empirical survey sites.

occupancy is given in Fig. 3. The colonization function we used in subsequent simulations is simply that any unoccupied cell within three cells (horizontally, vertically, and diagonally) of an occupied cell will be occupied during the next time step.

The procedure for determining the effects of hurricanes on occupancy was similar to the procedure used to determine a colonization function. The effect of hurricanes on occupancy was systematically varied in during a model development phase. We searched for hurricane effects that produced a good fit between simulated occupancy in cells corresponding to survey sites and the empirical estimates of post-hurricane occupancy (Fig. 3). The hurricane survival rate used in post-development simulations was 0.05 for all cells except for the cells located in Group 3 and Group 9 of Fig. 2. Cells located in Group 3 and Group 9 were given a hurricane survival probability of 0 because they are close to the ocean and have poorly formed dunes. These areas, therefore, experience greater flooding from storm surge. Although familiarity with the landscape influenced our decisions about how to simulate hurricane effects, the match between simulated and observed occupancy was the final criterion.

Hurricane return interval was modeled as a Poisson random variable with a fixed rate parameter. The simulation begins at a (uniform) random point between hurricanes. After a hurricane occurs, a new Poisson random interval is drawn and the occupancy simulation proceeds without a hurricane for the duration of the newly specified interval. Survival from the first to second year post-hurricane is computed from the corresponding survival map. Our second survival map is used to simulate survival from the second to the third year after a hurricane. Survival for all years after the third post-hurricane year is simulated with the third survival map. We believe this is reasonable because occupancy seems to asymptote by the fourth post-hurricane year (Fig. 4). This process is used to simulate occupancy dynamics for 100 years. At the end of the 100th year, we record whether or not at least one patch is occupied. This entire scenario is repeated 500 times, and the fraction of scenarios ending with 0 sites occupied is the probability of extinction.

We repeated scenarios with different hurricane return intervals in order to assess uncertainty in hurricane frequency. We varied the Poisson rate from 10 to 60 years in increments of 5 years. Modeling hurricanes frequency with consecutive Poisson intervals makes it extremely unlikely for hurricanes to occur in two consecutive years unless the intervals are unrealistically short (<5). Nonetheless, hurricanes Ivan and Katrina in 2004 and 2005, respectively affected *P. p. ammobates* (hurricane Dennis also impacted the area in 2005). Thus, the Poisson distribution may not fully capture observed hurricane frequency. We, therefore, ran additional simulations where the hurricane return interval is Poisson distributed with a mean rate of 15 years, but we imposed another hurricane the year after the fifth hurricane occurred.

We also assessed the impact of different levels of habitat loss on the probability of extinction. Habitat loss was incorporated by simply deleting a given fraction of randomly selected cells from the model before a simulation began. We explored habitat loss ranging from 0% to 50% of the original cells, at increments of 2.5%. All of the 500 replicates of a given hurricane regime had the same fraction of cells removed, but the identity of the cells was unique. Some spatial configurations of a given amount of habitat loss will likely produce probabilities of extinction that are greater or less than other spatial configurations of the same amount of habitat loss. The technique we employed averages out the effects of spatial configurations because we were interested the expected effect of habitat loss without regard to particular spatial configurations. This model could also be used to explore the effects of specific, hypothetical or proposed changes to the landscape on population viability. However, our goal is to understand the general effect of habitat loss without the becoming entangled in the associated issue spatial configurations.

3. Results

3.1. Dynamic occupancy model

Mean nightly detection probabilities (p) from 2006 to 2009 were 0.70, 0.55, 0.52, and 0.60. Detection probability over a

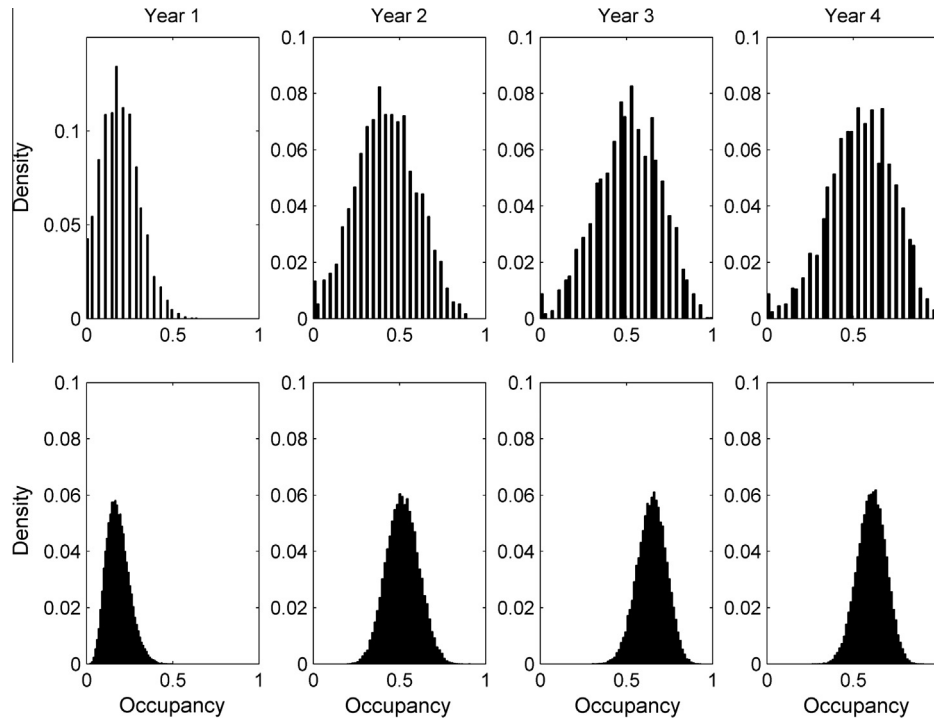


Fig. 3. Simulated occupancy rates for four years immediately after a hurricane (above) at locations where detection/non-detection data were collected, and occupancy rates estimated directly from the data (below). The extra variation in simulated occupancy reflects uncertainty in the post-hurricane recovery process.

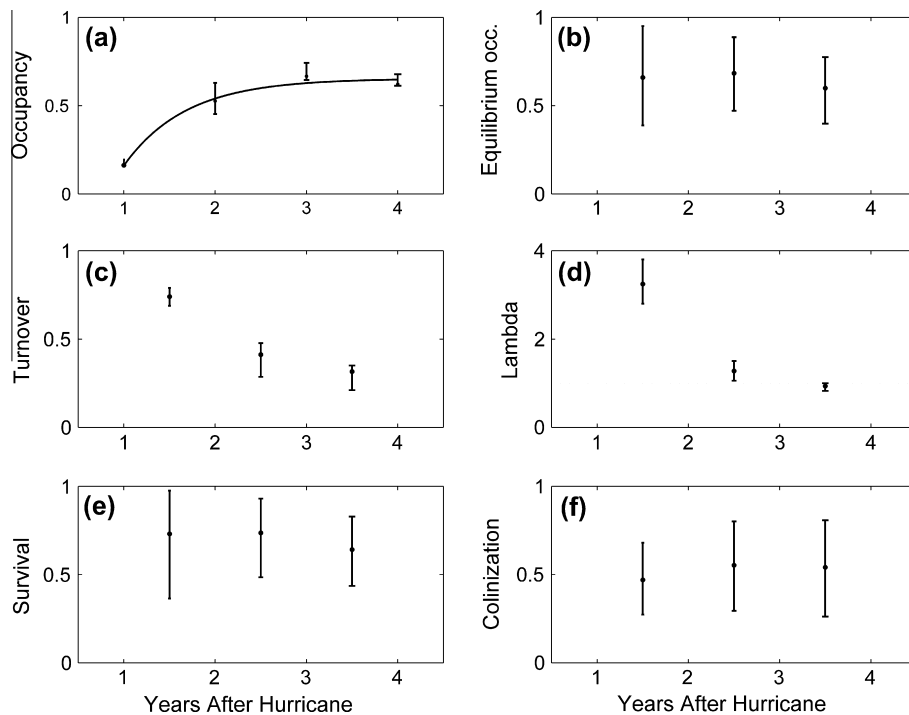


Fig. 4. Estimates of (a) occupancy, (b) equilibrium occupancy, (c) turnover, (d) occupancy growth rate, (e) survival, and (f) colonization of *Peromyscus polionotus ammobates* following catastrophic hurricanes. Error bars represent 95% Bayesian credible intervals. The curved line in (a) is the nonlinear regression fit to the four estimates of occupancy post hurricane. Processes that occur across consecutive years are plotted at the midpoint of the two relevant years.

five-night sampling period was, therefore, quite good. For example, in 2006 a typical five-night survey had a detection probability of $1 - 0.30^5 = 0.998$. The estimates of occupancy using data from sites that were sampled each summer from 2006 through 2009 increase at a decreasing rate (Fig. 4). The maximum occupancy probability predicted by the fit of a nonlinear regression to the estimates of

occupancy through time is 0.654, which closely matches the three estimates of equilibrium occupancy (ψ_y^*) derived from the hierarchical model (Fig. 4b). The estimated probability of turnover decreases through time at a decreasing rate (Fig. 4c), as does the estimates of population growth rate, λ (Fig. 4d). There is considerable uncertainty in survival (Fig. 4e) and colonization (Fig. 4f)

across years, so the initial recovery and subsequent equilibrium of occupancy cannot be easily decomposed into changes in survival and colonization through time.

The neural network correctly classified 100% of the sand in our validation dataset. The network incorrectly classified 1.5% of vegetation as “other”, and incorrectly classified 4.5% of “other” as sand. Fig. 5a and b provide an example of the network’s performance. Note that the pixels at the boundary between sand and vegetation are frequently misclassified as “other.” Also note in Fig. 5a,b that water on the asphalt parking lot was misclassified as vegetation. The particular location was chosen because it contains many cells from the three classes, and is not representative of the landscape.

The estimated posterior distributions of model-averaged effects for each covariate on survival rate are provided in Fig. 6. Model-averaged effects are calculated as the product $\omega\beta$, i.e. the influence of a covariate if it is included in a model ($\omega = 1$) or 0 if a variable is not included. When ω evaluates to 1, the magnitude of the covariate coefficient (β) is expressed. This can be seen in the histograms of Fig. 6 as lumps of probability mass on both sides of 0. The mean of the distributions in Fig. 6 were used to predict probability of survival in all 398 cells that comprise the landscape (Fig. 2).

3.2. Population simulations

The probability of extinction over 100 years is less than 0.05 for 85% of the combinations of habitat loss and hurricane frequency that we explored (Fig. 7). Even at the highest hurricane frequency, a 10-year mean return interval, the probability of extinction is less than 0.05 if habitat loss remains less than 15%.

Increasing hurricane frequency and increasing habitat loss both increase the probability of extinction nonlinearly. There is a threshold toward the more severe end of the parameter space that we explored where extinction probability changes very abruptly (Fig. 7). An interaction between hurricane frequency and habitat loss may be visualized in Fig. 7 as curvature in the isobars of extinction probability. Fig. 7 suggests that hurricane frequency and habitat loss mildly interact when the probability of extinction is low, but the interaction seems to become less pronounced as extinction risk approaches about 0.2. Simulation uncertainty can be seen as jaggedness in the lines plotted in Fig. 7. As others have noted (White et al., 2014), there is no reason to perform statistical tests of model output.

When hurricane return interval is 15 years and 40% of habitat is lost, then the probability of extinction is 0.166. If the scenario is then altered such that a hurricane immediately follows the fifth hurricane, then the probability of extinction nearly triples (0.458). In order to determine how much of this increase is attributable to the consecutiveness of the additional hurricane versus the simple addition of a hurricane to the 100 year period, we first note that, on average, there are 7.18 hurricanes in a 100 year period when mean return interval is 15 years and a sixth hurricane is made to immediately follow the fifth hurricane. If a consecutive hurricane is not imposed, then the mean hurricane return interval must be 13.9 years in order to obtain 7.18 hurricanes in a 100 year period. When the mean hurricane return interval is 13.9 years and 40% of habitat is lost, then the probability of extinction is 0.198. Thus, 89% of the increase in extinction probability from 0.166 to 0.458 can be attributed to consecutiveness of hurricanes while the remaining 11% is attributable to the increased number of hurricanes.

4. Discussion

Estimates of occupancy, turnover, and meta-population growth rate (λ) all indicate a remarkably smooth recovery of the

Alabama beach mouse during the four years subsequent to catastrophic hurricanes. Although we saw asymptotic properties of the recovery process after just three years, we also note that our occupancy-scale estimates of recovery are not necessarily identical to the recovery of population abundance. We emphasize that occupancy and abundance are different, yet related, measures of population status. The Alabama beach mouse has historically reached much higher densities than we observed (Rave and Holler, 1992), which strongly suggests that we did not witness the end of recovery of population size. Swilling et al. (1998) noted that a decade-long increase in abundance among many beach mouse subspecies could reflect recovery from Hurricane Elena in 1985, further suggesting that population size continues to grow after occupancy slows or stops. Royle and Kéry (2007) note that occupancy estimates of Cerulean Warblers (*Dendroica cerulea*) lag behind declines in abundance, a process they call the “inertia” of occupancy with respect to abundance. We note that the reverse is also possible when a species is increasing in abundance: If there

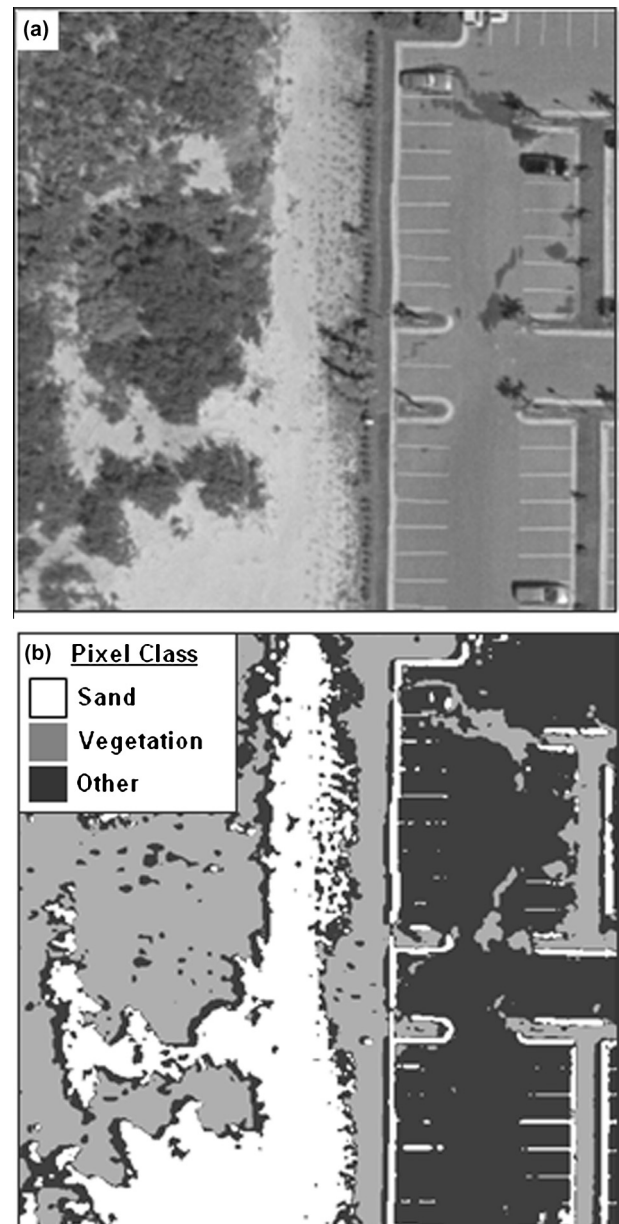


Fig. 5. Aerial photograph (a) and associated classification of pixels (b) into sand, vegetation, or “other”.

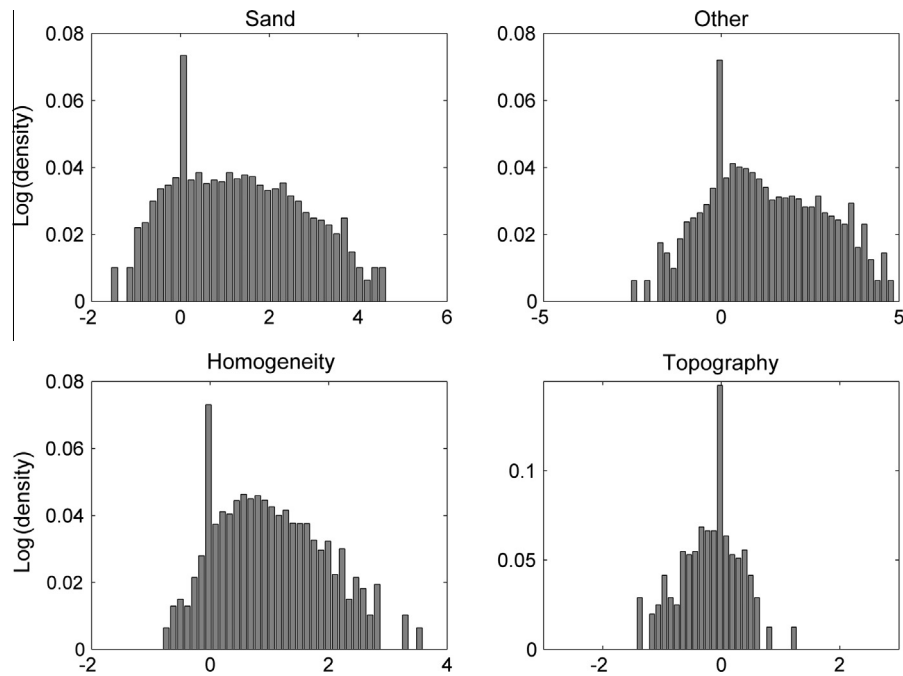


Fig. 6. Posterior distribution of effects of covariates ($\omega * \beta$) on survival. The x-axis is the effect of the covariate on the logit scale.

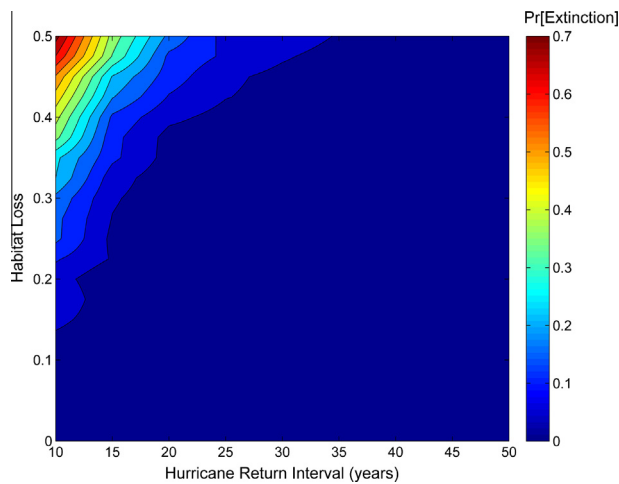


Fig. 7. The probability of extinction over a 100 year period for different levels of habitat loss and hurricane return interval.

are density-dependent effects on habitat quality and individuals are able to select optimal habitats (Fretwell and Lucas, 1970), then initial increases in population density will result in relatively large increases in occupancy. It is unclear if the relatively rapid recovery we witnessed is entirely related to occupancy vs. abundance metrics. If there is considerable variation in the speed of recovery events and we happened to witness a relatively quick recovery, then the recoveries we simulated in the PVA would be biased, and actual extinction risk could be higher.

Despite considerable uncertainty in all three estimates of equilibrium occupancy (Fig. 4b), the means of the posterior distributions are similar to one another and very closely match the estimated asymptote from the nonlinear regression model fitted to the four estimates of occupancy through time (Fig. 4a). This suggests that two survey seasons can yield a reliable estimate of equilibrium occupancy, but longer-lived species with slower demographic rates will require more than one year between survey periods.

Covariate information from all 61 sites was used to make spatial predictions of colonization and survival rates. In Fig. 6, the height of the bar at 0 indicates the probability that $\omega_j = 0$. The taller the bar at 0, the less likely it is that the given covariate meaningfully contributes to survival. We did not find a significant influence of the prior distribution of model weights [$\omega \sim \text{Bernoulli}(0.5)$] on our results. When we re-ran the analysis with lower priors [$\omega \sim \text{Bernoulli}(0.1)$], we obtained essentially the same results (not shown). Thus, the contribution of the covariates is not over-emphasized by our prior neutrality toward the meaningfulness of each covariate. Nonetheless, we acknowledge that the covariates weakly predict survival.

We used a spatially-explicit approach to simulating population dynamics of occupancy parameters estimated from detection/non-detection data. The salient components of our approach are: (i) survival is mapped across the entire area of interest, (ii) the spatial resolution of survival mapping matches the scale of data collection, (iii) the degree of spatial independence of survival is also mapped, (iv) colonization is a function of occupancy of neighboring patches (i.e. colonization is also spatially-explicit) (v) colonization rate is determined by searching for an algorithm that yields an occupancy rate at survey locations that is consistent with empirical data. Theoretically, multiple colonization algorithms can reproduce the occupancy dynamics at cells corresponding to where data were collected. However, we found only one combination of colonization and hurricane effects that came close to matching our empirical occupancy estimate. It seems unlikely to us that we could re-create the empirical occupancy estimate with different, yet reasonable, parameterizations of our model. The correspondence between simulated occupancy and observed occupancy also supports our map of spatial independence of survival (Fig. 2a).

Concern about the accuracy of PVA models typically focuses on (i) data availability and quality, (ii) model misspecification, and (iii) weak ability to validate predictions. Weak ability to validate predictions raises philosophical issues that stand apart from any particular population viability analysis. In the first quantitative validation of PVA concepts, Belovsky et al. (2002) conclude that the accuracy of PVA is critically sensitive to (i) and (ii), and that qualitative prediction is generally less problematic than quantitative

predictions of long-term survival probability. Similarly, Bessinger and Westphal (1998) and Allendorf and Ryman (2002) indicate that PVA is better suited to relative risk assessments of different future scenarios than absolute estimation of extinction probability. Likewise, we believe our assessment of the effects hurricanes and habitat loss and their interaction on extinction risk are more robust than our quantitative estimates of extinction risk. The accuracy of these estimates depends on the appropriateness of the assumptions we used to link our detection/nondetection data to the population simulation.

Our results suggest that the probability of extinction over 100 years may be rather low (<0.05) if habitat loss does not continue. Oli et al. (2001) used demographic data from before and after hurricane Opal (1995) and concluded that the extinction probability over 100 years for a single “population” of *P. p. ammobates* located in the Perdue Unit of the Bon Secour National Wildlife refuge is 0.358. This may not be inconsistent with our findings. Our extinction probability is a species-wide estimate whereas the extinction probability provided by Oli et al. (2001) pertains to a small fraction of the recently occupied geographic range of *P. p. ammobates*. Furthermore, we observed extremely low densities of *P. p. ammobates* after hurricane Katrina within the area of Oli et al.'s (2001) scope of inference, which qualitatively supports their finding that the area is susceptible to hurricane-induced extirpation. Notably, this area is within the largest, most pristine tract of remaining habitat. Although our estimates of extinction risk may be generally low, we emphasize that relative extinction risk is a more robust PVA output than estimates of absolute extinction risk. We found that extinction risk increases dramatically when consecutive hurricanes are simulated. This is a critical uncertainty. If consecutive, intense hurricanes, as experienced in 2004–2005, occur multiple times in a 100 year period, then *P. p. ammobates* faces a substantial risk of extinction. This extinction risk is further compounded by continued habitat loss.

The barrier island where *P. p. ammobates* occurs may not be older than 4.5 thousand years (Smith, 1986), and genetic work on coat coloration of *P. polionotus* (Hoekstra et al., 2006) suggests that *P. p. ammobates* immigrated from the mainland more recently than other *P. polionotus* subspecies. Even if *P. p. ammobates* had a long evolutionary history of surviving hurricanes, there is little reason to assume that the future will resemble the past. Sixty-one percent of the historically available habitat has already been compromised by human development (USFWS, 2005), and further habitat loss continues today. Furthermore, global climate change is expected by some climatologists to increase hurricane frequency and intensity (Webster et al., 2005; Trenberth, 2005), and our results suggest that the probability of extinction can increase dramatically if hurricanes occur more consecutively.

Author contributions

BJD originally formulated the idea. BJD and MRF developed methodology. MRF analyzed data, wrote computer code, and wrote most of the manuscript.

Acknowledgments

This work was supported by a grant from U.S. Fish and Wildlife Service to BJD. We profited from discussion with many USFWS staff, including Rob Tawes, Darren LeBlanc, Sandra Sneckenberger, and Jeremy Phillips. Three anonymous reviewers provided excellent commentary that helped us refine this manuscript. Adam Anderson, Garrett Davis and Eric Schroder provided excellent assistance with field work.

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